

Lab Assignment 3

Lab Assignment 3 – GridWorld

1. Write Python code to perform one iteration of policy evaluation for 3x3 GridWorld.
2. Write Python code to calculate the value of the specified state using the given policy for the 3×3 GridWorld.
3. Write Python code to determine the best action for the given state using value iteration.
4. Write Python code to find the shortest path from the Start state to the Goal state using the given optimal policy.
5. Write Python code to compare two paths based on path length and total reward.

Note: Use below data to write the codes for the above mentioned questions.

Given 3×3 GridWorld:

S		
	X	
		G

S = Start, G = Goal,

X = Blocked cell.

Valid actions: Up ($\uparrow$), Down ($\downarrow$), Left ($\leftarrow$), Right ($\rightarrow$).

Given Parameters:

Parameter	Value
Grid size	3×3
Step reward	-1
Goal reward	+10
Discount factor (γ)	0.9
Initial value $V(s)$	0 for all states